Decision Tree

Thursday, February 22, 2024 9:00 PM

ML

- Linear Regression
- Logistic Regression
- KNN →
- Decision Tree

DS. JIO

Task 1



97% Leave

Find the chance of leaving the Company

Classification of Model

Task 2

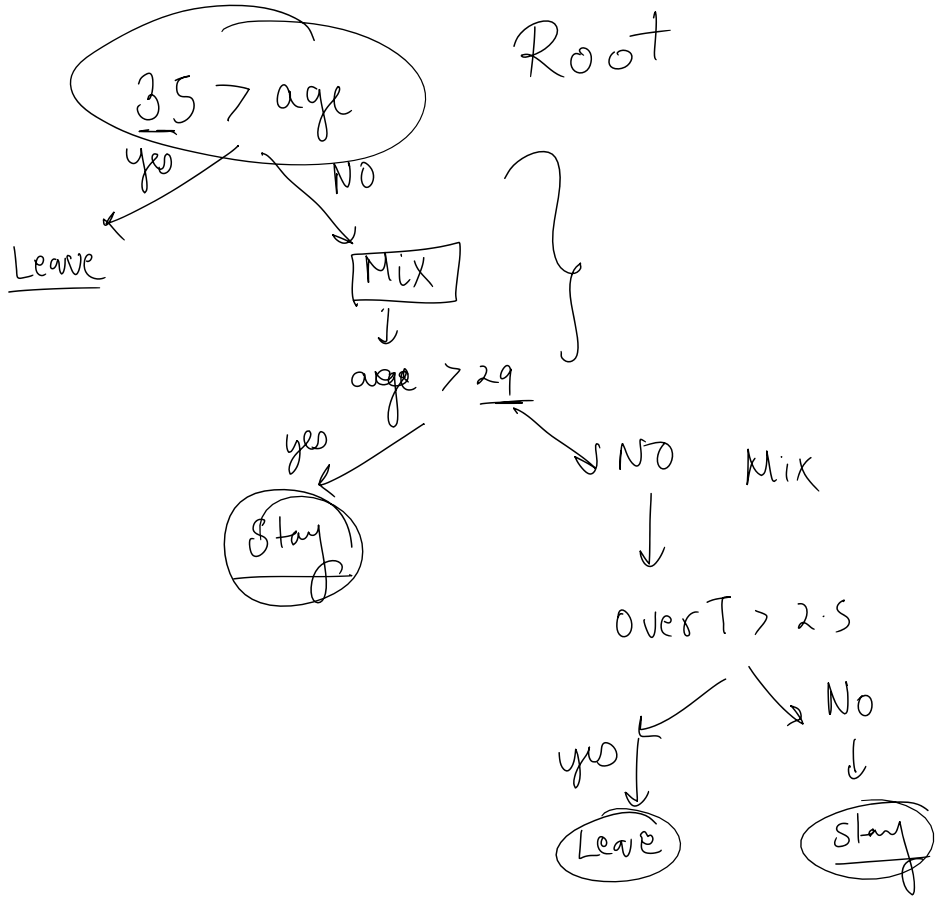
Find the factors for attrition of employee

Model

→ Inter predability

Decision Tree

- Series of question
- IMP question on Top



Data Non linear
Interpretability
} DT

Statement I : Topmost node is called leaf node

Statement II: Topmost node is called root node.

Statement III: Bottom most node are called

leaf nodes.

Statement IV: Nodes in between root and leaf are called decision nodes/ internal nodes.

Which of the following statements are true?

From <https://www.edgar.com/instructor/meetings/i/ml-decision-trees-1-27/>

Step — DT

① Find the Root

Question

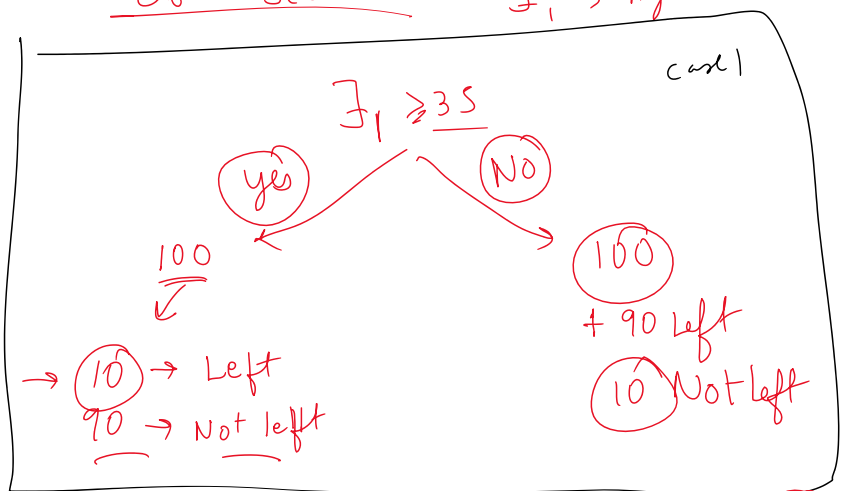
which question to put on Top?

Best question → Pure answer

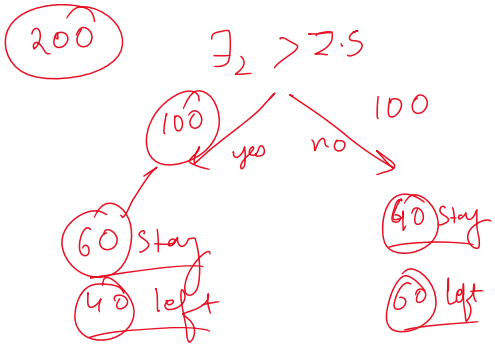
n = 200 → 100 (+) Leave

→ 100 (−) Not Leave

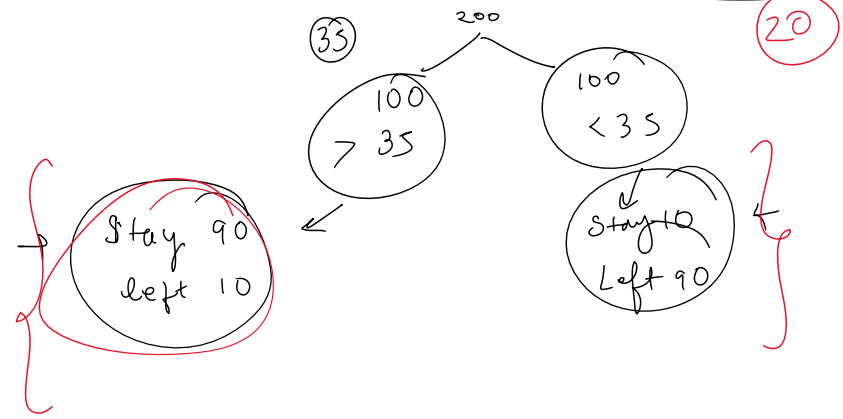
36 → Features J_1 → Age



J_2 — col OVERTIME



90 → Not left



20

Age → 71

Pure → Any 1 class is Dominating

(1)

{ 90% Leave
10% Not Leave }

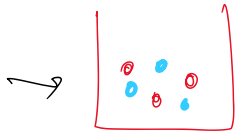
60 Leave
40 Leave Not

Pure

$$-\sum_{i=1}^{K} p(y_i) \log_2(y_i)$$

P → Class 1 } Leave
P → Class 0 } Stay

$$-\left[P(1)\log_2 P(1) + P(0)\log_2 P(0)\right]$$

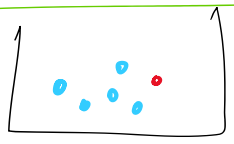


50% 50%

Red – 1
Blue – 0

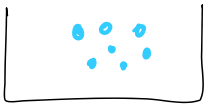
High Impure

$$-\left[\frac{3}{6}\log_2\frac{3}{6} + \frac{3}{6}\log_2\frac{3}{6}\right] \Rightarrow 1$$



Medium Impure

$$= -\left[\frac{5}{6}\log_2\frac{5}{6} + \frac{1}{6}\log_2\frac{1}{6}\right] \Rightarrow 0.65$$



Lowest

$$= \frac{6}{6}\log_2\frac{6}{6} + \frac{0}{6}\log_2\frac{0}{6} \Rightarrow 0$$

Entropy (Impure)

$n = 100 \rightarrow y \begin{cases} 0 & \text{Stay } 60 \\ 1 & \text{Leave } 40 \end{cases}$

$C_1 \rightarrow$ Gender
$C_2 \rightarrow$ Age < 35

$-\sum p(y) \log p(y)$

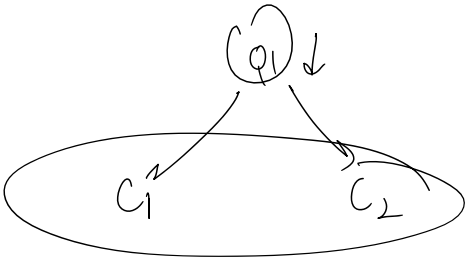
60 → +
40 → −

$-[0.6 \log_2 0.6 + 0.4 \log_2 0.4]$

$= 0.97$ (high)

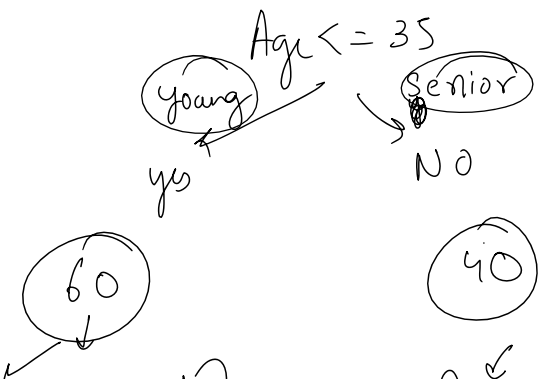
entropy.

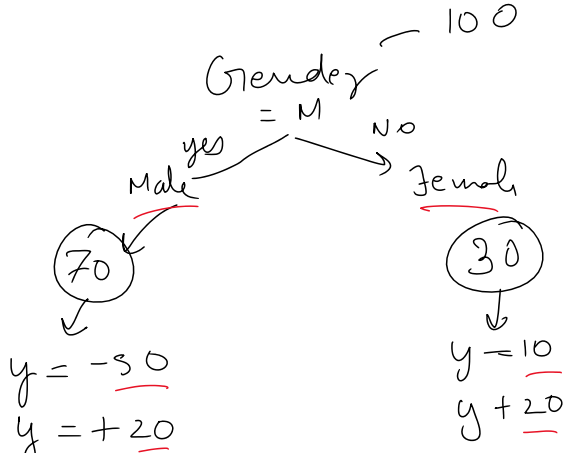
Parent

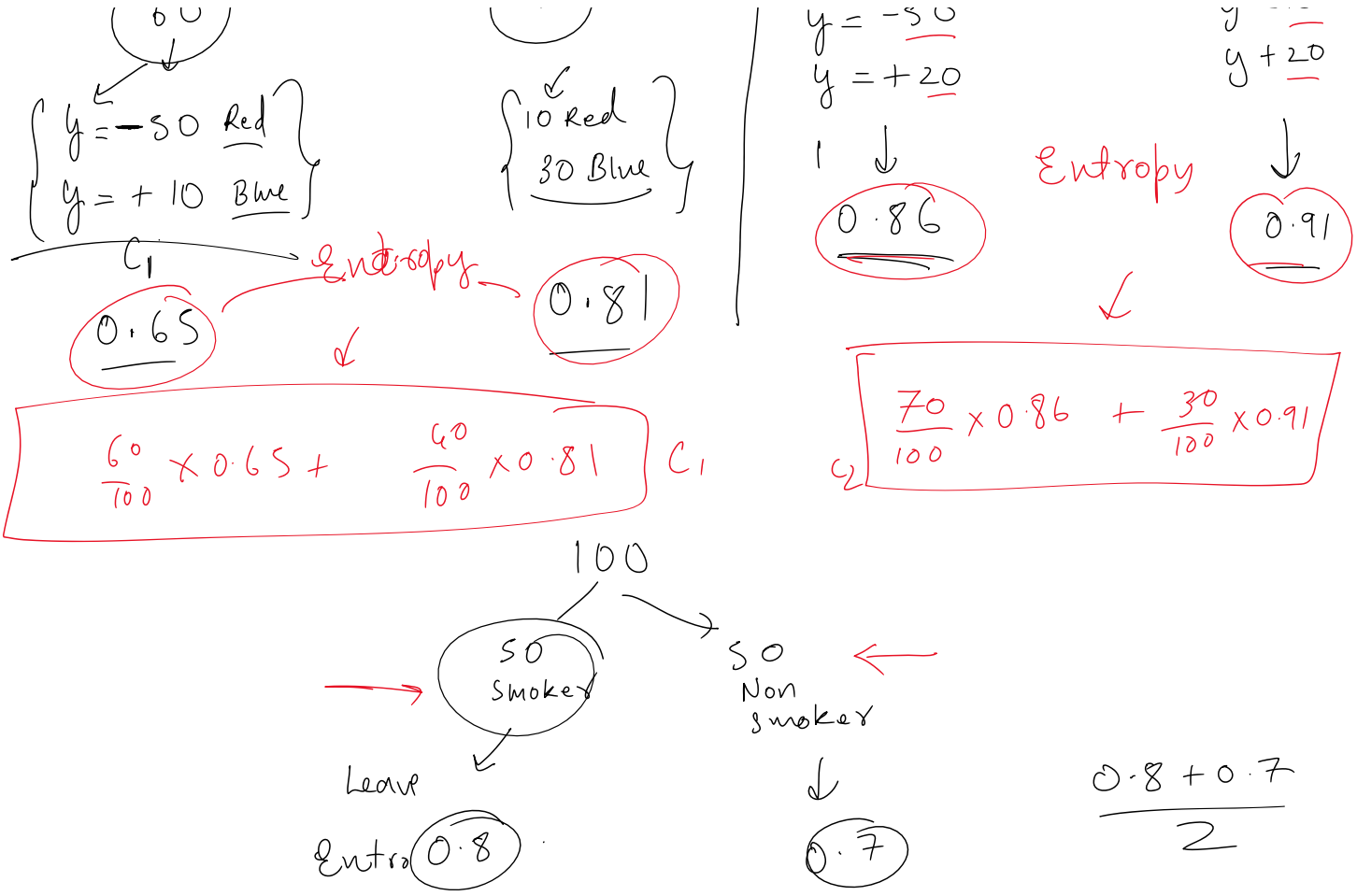


$T_1 \rightarrow$ Age < 35

$H[\text{Parent}] = 0.97$

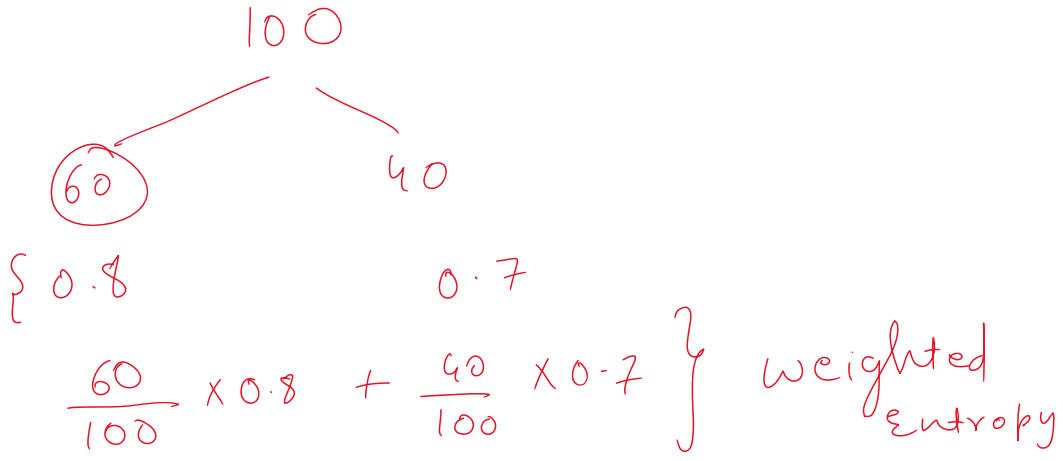






$\frac{0.8 + 0.7}{2}$

60



Parent 0.97

Parent 0.97

Combine C_1 → 0.1

C_2 → 0.9

case

0.97 − 0.1

0.87

Information Gain

0.97 − 0.9

IG_1 0.07

↑ IG_1 is how much Inf f_1 has provide to DT

→ {LSH} | {weighted KNN}

M → 70
F → 30
} 0.6